

✨HackCelestial 3.0 Midnight Task✨

Weather-Driven Digital Twin Enhancement for Hospitality & Travel Solutions

Weather can create complex, delayed, and cascading effects across Hospitality & Travel, influencing transportation, traveler behavior, hotel occupancy, attraction demand, restaurant demand, events, workforce availability, and resource utilization. **As an enhancement to their existing HackCelestial solution, teams are required to develop and integrate an AI-driven Digital Twin capability into their current system rather than build a separate standalone application.** The integrated Digital Twin should represent the relevant real-world entities, resources, relationships, and operational conditions already addressed by the team's solution and continuously estimate how they may behave under changing weather conditions. Using weather observations and forecasts together with the team's existing data, the model should learn the relationship between environmental conditions and the behavior of the solution's target ecosystem, identify direct and cascading effects, and generate a continuously evolving virtual representation of the affected system. Teams should enable their existing solution to simulate **what-if and counterfactual scenarios**, such as changes in rainfall intensity, temperature, storm duration, flooding, extreme heat, or other weather conditions, and estimate how these changes could alter demand, capacity, movement, availability, operations, or user behavior within their specific problem domain. The Digital Twin should be capable of updating its simulated state as new real-world data becomes available, propagating changes across interconnected entities, identifying secondary and higher-order effects, and providing probabilistic predictions with associated uncertainty. Teams should demonstrate how their existing solution behaves under both normal and extreme weather scenarios and how the integrated AI model can simulate alternative future states without affecting the actual system. The challenge is therefore **not to create a new weather application or standalone Digital Twin**, but to extend the team's existing Hospitality & Travel solution with an intelligent, continuously learning simulation layer that can understand how weather-driven changes propagate through the system and use those insights to improve its existing functionality.

Mandatory Integration Requirements

As part of this additional task, the enhanced solution **must include**:

1. **Live Weather Integration:** Integrate a real-time weather API and use live/current or forecast weather data as an input to the AI model.
2. **Geospatial Map Visualization:** Provide a map-based visualization showing the relevant locations, entities, weather conditions, and/or simulated impact propagation within the team's existing solution.



3. **Real-World Social Signal Integration:** Integrate relevant social-media or publicly available social signals to capture real-world traveler/user reactions, reports, trends, or emerging conditions related to the weather event.
4. **Digital Twin What-If Simulation:** Demonstrate at least one interactive scenario where changing a weather parameter (such as intensity, duration, location, or temperature) produces a corresponding change in the team's existing system through the Digital Twin.